Agentic AI Water Distribution Leak Detection & Resource Optimization System

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Water Network Decision-Support Report

This report summarizes sensor monitoring, anomaly detection, leak assessment, water-loss analysis, maintenance prioritization, and operational recommendations.

1. Network Monitoring Summary

Metric	Value
Anomalous sensor records	12
Distribution zones with anomalies	3
Human approval	Required
Physical infrastructure control	Not automatic

2. Flow & Pressure Anomaly Analysis

Timestamp	DMA	Flow Z	Pressure Z	Nighttime
2026-09-10 22:00:00	DMA-A	-0.15	-3.02	False
2026-09-12 14:00:00	DMA-A	0.09	3.07	False
2026-09-20 02:00:00	DMA-B	0.42	-9.62	True
2026-09-20 03:00:00	DMA-B	0.50	-8.65	True
2026-09-20 04:00:00	DMA-B	0.22	-10.26	True
2026-09-20 05:00:00	DMA-B	0.68	-8.81	True
2026-09-20 06:00:00	DMA-B	2.77	-9.06	False
2026-09-25 01:00:00	DMA-C	0.23	-7.65	True
2026-09-25 02:00:00	DMA-C	0.49	-8.68	True
2026-09-25 03:00:00	DMA-C	0.35	-7.47	True
2026-09-25 04:00:00	DMA-C	1.09	-8.49	True
2026-09-25 05:00:00	DMA-C	0.28	-8.77	True

3. Leak Detection & Localization

Total leak assessments: 12
High-confidence assessments: 9
Localization approach: Suspected DMA / pipe section based on available topology and sensor evidence.
Exact underground location: Not claimed.

DMA	Pipe	Score	Confidence	Status
DMA-A	Pipe-A1	70.00	Medium	Investigation Required
DMA-A	Pipe-A1	70.00	Medium	Investigation Required
DMA-B	Pipe-B1	100.00	High	Investigation Required
DMA-B	Pipe-B1	100.00	High	Investigation Required
DMA-B	Pipe-B1	100.00	High	Investigation Required

DMA	Pipe	Score	Confidence	Status
DMA-B	Pipe-B1	100.00	High	Investigation Required
DMA-B	Pipe-B1	70.00	Medium	Investigation Required
DMA-C	Pipe-C1	100.00	High	Investigation Required
DMA-C	Pipe-C1	100.00	High	Investigation Required
DMA-C	Pipe-C1	100.00	High	Investigation Required
DMA-C	Pipe-C1	100.00	High	Investigation Required
DMA-C	Pipe-C1	100.00	High	Investigation Required

4. Water Balance & Loss Analysis

DMA	Supplied	Consumption	Unexplained	Loss %	High Conf.
DMA-A	84,465	65,430	19,035	22.54%	0
DMA-B	94,517	73,031	21,487	22.73%	4
DMA-C	74,842	57,672	17,171	22.94%	5

Important: Unexplained water does not automatically represent confirmed leakage. It may also include measurement issues, authorized/unmetered consumption, or other network conditions.

5. Maintenance & Field Resource Planning

DMA	Pipe	Priority	Score	Team	Approval
DMA-C	Pipe-C1	Priority 1 - Critical	92.94	Field Team Alpha	Pending Human Approval
DMA-B	Pipe-B1	Priority 1 - Critical	88.73	Field Team Beta	Pending Human Approval
DMA-A	Pipe-A1	Priority 3 - Medium	53.54	Field Team Gamma	Pending Human Approval

6. Human-in-the-Loop Operations

Approval status: Pending Human Approval
Execution mode: Manual / operator-controlled
Safety: The prototype does not autonomously operate pumps, valves, or other physical infrastructure.

Recommended field actions should be reviewed by authorized water-utility personnel before execution.

7. Operational Summary

The system combines sensor monitoring, demand forecasting, flow and pressure anomaly detection, leak assessment, water balance analysis, maintenance prioritization, and operational recommendations into a human-approved decision-support workflow.